

RELATIVITY PROBLEMS

Q1. What will be the length of a meter rod appear to a person travelling parallel to the length of the rod at a speed of $0.8c$ relative to rod?

Given:- $l_0 = 1.0\text{m}$; $v = 0.8c$

Formula :- $l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$

Solution :- $l = 1.0 \sqrt{1 - 0.8^2} = 0.6\text{m}$

Ans:- The apparent length of the rod while in motion is 0.6m.

Q2. A meter ruler moves past an observer on the earth with a velocity of 2.5×10^{10} cm/ sec, along the direction of its length. What is its apparent length with respect to the observer?

Given:- $l_0 = 1.0\text{m}$; $v = 2.5 \times 10^8 \text{m/s}$

Formula:- $l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$

Solution :- $l = 1.0 \sqrt{1 - \left(\frac{2.5 \times 10^8}{3 \times 10^8}\right)^2} = 55.27\text{cm}$

Ans:- The apparent length of the moving ruler is 55.27cm.

Q3. A spaceship 50m long passes the earth at a speed of 2.8×10^8 m/s. what will be its apparent length?

Given:- $l_0 = 50\text{m}$; $v = 2.8 \times 10^8 \text{m/s}$

Formula:- $l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$

Solution :- $l = 50 \sqrt{1 - \left(\frac{2.8 \times 10^8}{3 \times 10^8}\right)^2} = 18\text{m}$

Ans:- The apparent length of the moving spaceship is 18m.

Q4. A rocketship is 50m long. When it is on flight its length appears to be 49.5 to an observer on ground. Find the speed of the rocket?

Given:- $l = 49.5\text{m}$; $l_0 = 50\text{m}$

Formula:- $l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$

Solution:- $49.5 = 50 \sqrt{1 - \frac{v^2}{c^2}}$

$$1 - \frac{v^2}{c^2} = 0.9801$$

$$v = 0.141c$$

Ans:- The speed of the rocket is 0.141c .

Q5. A certain particle called meson has a life time $2 \times 10^{-6}\text{sec}$; a] What is the mean life time when the particle is travelling with a speed of $2.9994 \times 10^8 \text{ m/sec}$?; b] How far does it go during one mean life?

Given:- $t_0 = 2 \times 10^{-6}\text{sec}$; $v = 2.9994 \times 10^8 \text{ m/sec}$

Formula:- $t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$; distance (d) = Speed (v) x time (t)

Solution:- $t = \frac{2 \times 10^{-6}}{\sqrt{1 - \left(\frac{2.9994 \times 10^8}{3 \times 10^8}\right)^2}} = 31.63 \times 10^{-6} \text{ sec}$

Distance travelled by meson during mean life time,

$$d = (2.994 \times 10^8)(31.63 \times 10^{-6})$$

$$= 9470\text{m}$$

Ans:- The mean life time is $31.63 \times 10^{-6}\text{sec}$ and the distance is 9470m.

Q6. The mean life of meson is 2×10^{-8} sec. calculate the mean life of a meson moving with a velocity $0.8c$?

Given :- $t_0 = 2 \times 10^{-8}$; $v = 0.8c$

Formula:- $t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$

Solution:- $t = \frac{2 \times 10^{-8}}{\sqrt{1 - (0.8)^2}} = 3.3 \times 10^{-8} \text{ sec}$

Ans:- The mean life is 3.3×10^{-8} sec .

Q7. A wrist watch keeps correct time on earth. If it is worn by pilot in spaceship, leaving earth with constant velocity of 10^9 cm/sec. How many seconds does it appear to lose in one day with respect to the observer on the Earth.

Given:- $v = 10^9$ cm/sec $= 10^7$ m/sec; $t = 24$ hrs in a day

Formula:- $t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$

Solution:- $24 = \frac{t_0}{\sqrt{1 - \frac{1}{900}}}$

$$t_0 = 24 \left[1 - \frac{1}{900} \right]^{-1/2}$$

Here, $1/900$ is very small w.r.t 1 hence we can use the binomial expansion and neglect higher order terms

$$t_0 = 24 \left[1 - \frac{1}{1800} \right] = 24 - \frac{1}{75} \text{ hrs}$$

$$\text{hence loss in 24 hours} = \frac{1}{75} \text{ hrs} = 48 \text{ sec}$$

Ans:- wrist watch loses 48 sec in a day with respect to observer on earth.